CS 210: Worksheet 11 *Dr. Mudassir Shabbir, October 1, 2026*

Student ID 1: _____ **Seat Location 1:** row:_____ col:_____

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Sets and set operations. Distinguish carefully between $\in$ and $\subseteq$. To prove $X = Y$, show $X \subseteq Y$ and $Y \subseteq X$ by fixing an arbitrary element. To disprove a claim about all sets, give one specific counterexample and check it.

1. *Easy*

Let $T = \{0,\ \{0\},\ \{1\}\}$. Mark each statement true (T) or false (F).

$$\{0\} \in T \qquad \{0\} \subseteq T \qquad \{1\} \subseteq T \qquad \varnothing \in T \qquad \varnothing \subseteq T$$

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2. *Easy*

Let $R = \{1,\ \{1\}\}$.

(a) Find $|R|$ and $|\mathcal{P}(R)|$.

(b) List $\mathcal{P}(R)$.

(c) Is $\{1\} \in \mathcal{P}(R)$? Is $\{\{1\}\} \in \mathcal{P}(R)$?

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3. *Easy*

Let $U = \{1, \ldots, 8\}$, $A = \{1, 2, 3, 6\}$, and $B = \{2, 4, 6, 8\}$.

(a) Find $A \cup B$, $A \cap B$, $B \setminus A$, and $(A \cup B)^c$.

(b) Write the bit strings for A and B, in the order $1, \ldots, 8$.

(c) Compute the bitwise AND and XOR of these strings. Decode both.

4. *Moderate*

Decide. Then give a proof or one checked counterexample.

(a) If $A \setminus C = B \setminus C$, must $A = B$?

(b) If $A \subseteq B$, must $\mathcal{P}(A) \subseteq \mathcal{P}(B)$?

(c) If $A \triangle C = B \triangle C$, must $A = B$?

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5. *Easy (guided)*

Prove that for all sets A, B, and C,

$$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C).$$

(a) Write $x \in A \setminus (B \cap C)$ as a logical statement about $x \in A$, $x \in B$, and $x \in C$. Simplify it using De Morgan's law.

(b) Show $A \setminus (B \cap C) \subseteq (A \setminus B) \cup (A \setminus C)$.

(c) Show $(A \setminus B) \cup (A \setminus C) \subseteq A \setminus (B \cap C)$.

(d) Confirm the identity by completing the membership table.

A	B	C	$B \cap C$	$A \setminus (B \cap C)$	$A \setminus B$	$A \setminus C$	$(A \setminus B) \cup (A \setminus C)$
1	1	1					
1	1	0					
1	0	1					
1	0	0					
0	1	1					
0	1	0					
0	0	1					
0	0	0					

6. *Moderate*

In $\{1, \ldots, 100\}$, let D_m be the set of multiples of m.

(a) How many are divisible by 2, 3, or 5?

(b) How many are divisible by exactly one of them?

Hint: $|D_m| = \lfloor 100/m \rfloor$ and $D_2 \cap D_3 = D_6$.

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7. *Moderate*

Let D_m be the set of integers in $\{1, 2, \ldots, 1000\}$ that are divisible by m.

(a) Use inclusion–exclusion to find $|D_3 \cup D_5 \cup D_7|$.

(b) How many of these integers are divisible by *exactly one* of 3, 5, and 7?

Hint: $D_3 \cap D_5 = D_{15}$, and $|D_m| = \lfloor 1000/m \rfloor$.

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8. *Moderate*

Let $A = \{2r : r \in \mathbb{Z}\}$ and $B = \{2s - 2 : s \in \mathbb{Z}\}$. Is $A = B$? Prove your answer.
Hint: $A = B \iff A \subseteq B \ \wedge \ B \subseteq A$.

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9. *Easy*

Available permissions: camera, location, microphone. Any subset is allowed as a bundle.

(a) How many bundles?

(b) How many contain camera?

10. *Easy*

Let $U = \mathbb{R}$, $A = (-1, 0]$, and $B = [0, 1)$. Find

$$A \cup B, \qquad A \cap B, \qquad B \setminus A, \qquad A^c.$$

11. *Moderate*

Let $A_i = (-1/i, 1/i)$ for $i = 1, 2, 3, \ldots$. Find

$$A_1 \cup A_2 \cup A_3, \qquad A_1 \cap A_2 \cap A_3, \qquad \bigcup_{i=1}^{\infty} A_i, \qquad \bigcap_{i=1}^{\infty} A_i.$$

12. *Easy*

P: Potions. D: Defence Against the Dark Arts. C: Charms. How many students take at least one class?

Individual classes	Pairwise overlaps	All three
$\lvert P\rvert = 15$	$\lvert P \cap D\rvert = 6$	$\lvert P \cap D \cap C\rvert = 4$
$\lvert D\rvert = 11$	$\lvert P \cap C\rvert = 8$	
$\lvert C\rvert = 13$	$\lvert D \cap C\rvert = 7$	